

révision finale de chapitre 01

1/ The floor function:

A/ Properties:

- $[x] \in \mathbb{Z}$
- $[x+m] = [x] + m$
- $[x] \leq x < [x] + 1$
- $x - 1 < [x] \leq x$

EX:

$$[\pi] = [3.14] = 3$$

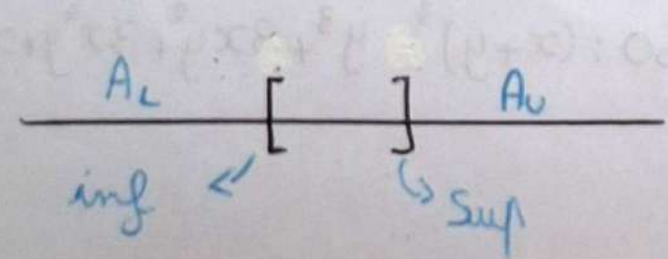
$$[-6.7] = -7$$

$$[0.7] = 0$$

2/ Lower bounds and upper bounds

- 1/ bounded above $x \leq M$
- 2/ bounded below $x \geq m$
- 3/ bounded $m \leq x \leq M$

3/ Sup, inf, min, Max



$\Rightarrow$ if $A \in I \Rightarrow \max = A$ Sinon!
 $A \notin I \Rightarrow \max \nexists$

$\Rightarrow$ if $B \in I \Rightarrow \min = B$ Sinon!
 $B \notin I \Rightarrow \min \nexists$

4/ Characterization of Sup and inf

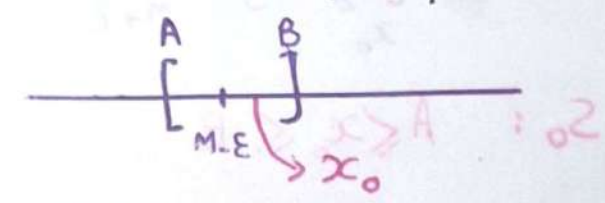
$$\text{Sup } A = M / \max \nexists$$

$$M = \text{Sup } A \Leftrightarrow \begin{cases} \forall x \in A, x \leq M \\ \forall \epsilon > 0, \exists x \in A, x > M - \epsilon \end{cases}$$

dans les intervals:

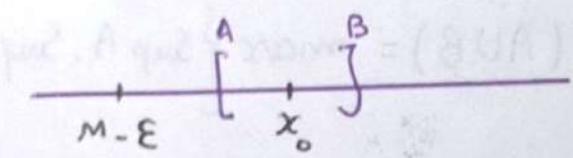
we have two cases:

Case 01: $M - \epsilon \gg \inf A$



So: $M - \epsilon < x_0 \leq B$

Case 02: $M - \epsilon < \inf A$



So: $A \leq x_0 \leq B$

intervale : $[a, b], \forall z \in \mathbb{R},$
 $a < z < b$
 $z \in [a, b]$

$\mathbb{N}, \mathbb{Z}, \mathbb{Q} \Rightarrow$ not an interval

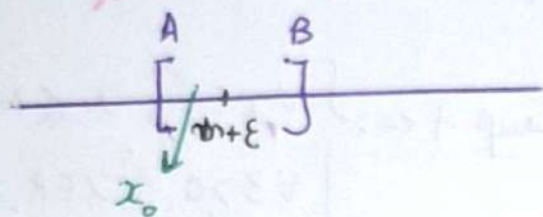
$$\inf A = m / \min \exists$$

$$m = \inf A \Leftrightarrow \begin{cases} \forall x \in A; x \geq m \\ \forall \varepsilon > 0, x \in A, x < m + \varepsilon \end{cases}$$

dans les intervalles

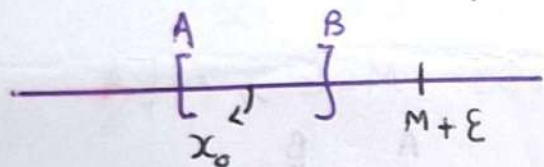
we have two cases:

Case 01: $m + \varepsilon \ll \sup A$



So: $A < x_0 < m + \varepsilon$

Case 02: $m + \varepsilon \geq \sup A$



So: $A \leq x_0 \leq B$

Properties

$$\sup (A \cup B) = \max \{ \sup A, \sup B \}$$

$$\inf (A \cup B) = \min \{ \inf A, \inf B \}$$

$$\sup (A + B) = \sup A + \sup B$$

$$\inf (A + B) = \inf A + \inf B$$

The density in $\mathbb{R}$

Between any two reals there exist a rational number that

$$\forall x \in \mathbb{R}, y \in \mathbb{R} (x < y) \Rightarrow \exists q \in \mathbb{Q},$$

$$x < q < y$$

Therefore: $\bar{\mathbb{Q}} = \mathbb{R}$

useful remarkable identities

Newton's Binomial theorem

$$(a+b)^n = \sum_{k=0}^n C_n^k \cdot x^k y^{n-k}$$

EX:

$$\begin{aligned} (x+y)^3 &= \sum_{k=0}^3 C_3^k \cdot x^k y^{3-k} \\ &= C_3^0 y^3 + C_3^1 x y^2 + C_3^2 x^2 y + C_3^3 x^3 y^0 \end{aligned}$$

and:

$$C_3^0 = \frac{3!}{0!(3-0)!} = 1$$

$$C_3^1 = \frac{3!}{1!(3-1)!} = \frac{1 \times 2 \times 3!}{1(2!)} = 3$$

$$C_3^2 = \frac{3!}{2!(3-2)!} = \frac{3!}{2!1!} = 3$$

$$C_3^3 = \frac{3!}{3!(3-3)!} = 1$$

So: $(x+y)^3 = y^3 + 3xy^2 + 3x^2y + x^3$

$$|x+y| \leq |x| + |y|$$

• if $x+y \geq 0 \Rightarrow |x+y| = x+y$

$\forall x \leq |x|$ donc $y \leq |y|$

$$x+y \leq |x| + |y|$$

• if $x+y < 0 \Rightarrow |x+y| = -x-y$

$-x \leq |x|$ donc $-y \leq |y|$

$$|x+y| \leq |x| + |y|$$

donc $|x+y| \leq |x| + |y|$

$$\frac{|x+y|}{1+|x+y|} \leq \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|}$$

on pose : $f(x) = \frac{x}{1+x}, x \geq 0$

$$f'(x) = \frac{1}{(1+x)^2} > 0$$

we have $|x+y| \leq |x| + |y|$

$$f(|x+y|) \leq f(|x|) + f(|y|)$$

$$\frac{|x+y|}{1+|x+y|} \leq \frac{|x|+|y|}{1+|x|+|y|}$$

$$\frac{|x+y|}{1+|x+y|} \leq \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|}$$

$$(a^n - b^n) = (x^{m-1} + x^{m-2}y + x^{m-3}y^2 + \dots + x^{m-n}y^{n-1}) \cdot x(x-y)$$

EX:

$$(x^2 - y^2) = (x-y)(x+y)$$

$$= (x-y)(x+y)$$

$$(x^4 - y^4) = (x-y)(x^3 + x^2y + xy^2 + y^3)$$

$$= (x-y)(x^3 + x^2y + xy^2 + y^3)$$

The absolute value

Proof:

$$|a| \cdot |b| = |a \cdot b|$$

Case 01: $a > 0; b > 0$

$$|a| \cdot |b| = a \cdot b$$

Case 02: $a \leq 0; b \leq 0$

$$|a| \cdot |b| = (-a) \cdot (-b) = ab = |ab|$$

Case 03: $a < 0; b > 0$

$$|a| \cdot |b| = (-a) \cdot (b) = -ab = |ab|$$

$$|a| \leq b \Leftrightarrow -b \leq a \leq b$$

$$|a| \leq b; -|a| \geq -b$$

and we know:

$$-b \leq -|a| \leq a \leq |a| \leq b$$

$$\Downarrow$$

$$-b \leq a \leq b$$

$$a \geq 0 \Rightarrow |a| = a \leq b$$

$$a \leq 0 \Rightarrow |a| = -a \geq -b$$

$$\text{So: } -a \leq b = |a|$$

$$\text{and } |a| \leq b \Leftrightarrow -b \leq a \leq b$$

The properties:

$$\bullet |xy| = |x| \cdot |y|$$

$$\bullet |x+y| \leq |x| + |y|$$

$$\bullet |x| \leq a \Leftrightarrow -a \leq x \leq a$$

$$\bullet |x| \geq a \Leftrightarrow x \geq a \vee x \leq -a$$

$$\bullet |x-y| = |x| \cdot |y|$$

$$\bullet ||x| - |y|| \leq |x-y|$$

$$\bullet -|x| \leq |x| \leq |x|$$

$$f(x+T) = f(x)$$

$$\sin(x+2\pi) = \sin(x)$$

$$\sin(ax) \Rightarrow T = \frac{2\pi}{a}$$

for ex:

$$\sin\left(\frac{2\pi}{7}x\right) \Rightarrow T = \frac{2\pi}{\frac{2\pi}{7}} = 2\pi \times \frac{7}{2\pi}$$

$$0 < 6 < T = 7$$

when we have a statement but with n for example

$$A = \left\{ (-1)^m + \frac{3}{5^m} ; m \in \mathbb{N}^* \right\}$$

$$A = B \cup C$$

The set B : when n is even

$$\Rightarrow A = 1 + \frac{3}{5^m}$$

$$2 \ll m < +\infty$$

$$10 \ll 5^m < +\infty$$

$$0 < \frac{1}{5^m} < \frac{1}{10}$$

$$1 < 1 + \frac{1}{5^m} < \frac{13}{10}$$

$$\text{So: } B =]1, \frac{13}{10}[$$

$$\sup B = \frac{13}{10}$$

$$; \inf B = 1 ; \max = \frac{13}{10}$$

$$A_0 = \left[\frac{13}{10}, +\infty[; A_0 =]-\infty, 1[; \min \nexists$$

construction of $\inf B = 1$

$$1 = \inf B \left\{ \forall x \in B ; x > 1 \right.$$

$$\left. \forall \varepsilon > 0, \exists x \in B, x < 1 + \varepsilon \right.$$

we have:

$$x_0 \in B \Rightarrow x_0 = 1 + \frac{3}{5^{m_0}} \text{ So:}$$

$$1 = \inf B \left\{ \forall m_0 \in B, 1 + \frac{3}{5^{m_0}} \right.$$

$$\left. \forall \varepsilon > 0, \exists m_0 > 2, 1 + \frac{3}{5^{m_0}} < 1 + \varepsilon \right.$$

we have:

$$1 + \frac{3}{5^{m_0}} < 1 + \varepsilon$$

$$m_0 > \frac{3}{5\varepsilon} \Rightarrow m_0 = \left\lceil \frac{3}{5\varepsilon} \right\rceil + 1$$

The set C : when n is odd

$$\Rightarrow A = -1 + \frac{3}{5^m}$$

$$1 \ll m < +\infty$$

$$10 \ll 5^m < +\infty$$

$$0 < \frac{1}{5^m} < \frac{1}{10}$$

$$-1 < -1 + \frac{3}{5^m} < -2/5$$

$$\text{So: } C =]-1, -2/5[$$

$$A_0 = \left[-\frac{2}{5}, +\infty[; \sup = -\frac{2}{5} ; \max = -\frac{2}{5}$$

$$A_0 =]-\infty, -1[; \inf = -1 ; \min \nexists$$

construction of $\inf C = -1$

$$-1 = \inf C \left\{ \forall x \in C, x > -1 \right.$$

$$\left. \forall \varepsilon > 0, \exists x \in C, x < -1 + \varepsilon \right.$$

$$\text{we have } x_0 \in C \Rightarrow x_0 = -1 + \frac{3}{5^{m_0}} \text{ So}$$

$$-1 = \inf C \left\{ \forall m_0 \in C, -1 + \frac{3}{5^{m_0}} \right.$$

$$\left. \forall \varepsilon > 0, \exists m_0 > 1, -1 + \frac{3}{5^{m_0}} < -1 + \varepsilon \right.$$

$$\text{we have } -1 + \frac{3}{5^{m_0}} < -1 + \varepsilon$$

$$m_0 > \frac{3}{5\varepsilon} \Rightarrow m_0 = \left\lceil \frac{3}{5\varepsilon} \right\rceil + 1$$

The set A :

$$\sup A = \max \left\{ \sup B, \sup C \right\} = \frac{13}{10}$$

$$\inf A = \min \left\{ \inf B, \inf C \right\} = -1$$

$$\max = \frac{13}{10} ; \min \nexists$$

The proof by induction:

$$i) \bullet \sum_{k=1}^n k! \cdot k = (n+1)! - 1$$

Step 01: for $n_0 = 1$ we have:

$$1! \cdot 1 = 1 \quad \text{So } P(n_0) \text{ is true}$$

$$(1+1)! - 1 = 1$$

Step 02: assume that $P(m)$ is true and we show that $P(m+1)$ is true

$$\left(\sum_{k=1}^{m+1} k! \cdot k = (m+2)! - 1 \right)$$

we have:

$$\sum_{k=1}^m k! \cdot k = (m+1)! - 1$$

$$\sum_{k=1}^{m+1} k! \cdot k = \underbrace{1 \times 1! + 2 \times 2! + \dots + (m!) \cdot m}_{P(m)} + ((m+1)! - 1)$$

$$k! \cdot k = (m+1)! - 1 + (m+1)! (m+1)$$

$$= (m+1)! (1 + m+1) - 1$$

$$= (m+1)! (m+2) - 1$$

$$= (m+2)! - 1 \quad \text{So } P(m+1) \text{ is true}$$

Therefore by induction $P(m)$ is true

$$ii) \sum_{k=2}^n \frac{1}{k} < 2\sqrt{n}$$

Step 01: for $n_0 = 1$ we have

$$\frac{1}{1} = 1 ; 2\sqrt{1} = 2/1 < 2$$

Step 02: assume that $P(m)$ is true

and we show that $P(m+1)$ is true

$$\sum_{k=2}^{m+1} \frac{1}{k} < 2\sqrt{m+1}$$

when we have equation

• $[\sqrt{x^2}] = 4$

• we know that $[x] \leq x \leq [x] + 1$

So: $[x] \leq x \leq [x] + 1$

$4 \leq x \leq 5$

$4^2 \leq x^2 \leq 5^2$

$4 \leq \sqrt{x^2} \leq 5$

$4 \leq |x| \leq 5$

$|x| \leq 5 \Rightarrow -5 \leq x \leq 5$

$|x| \geq 4 \Rightarrow x \in]-\infty, -4] \cup [4, +\infty[$

$S =]-5, -4] \cup [4, 5[$

• $[x[x]] = 1$

So: $[x[x]] \leq x[x] \leq [x[x]] + 1$

$1 \leq x[x] \leq 2$

we have:

$x \in]-\infty, -1$

$|x+2| - |x-3| + |x-4| > 0$

$\Rightarrow |x+2| \begin{cases} x+2, x+2 \geq 0 \Rightarrow x \geq -2 \\ -x-2, x+2 < 0 \Rightarrow x < -2 \end{cases}$

$|x-3| \begin{cases} x-3, x-3 \geq 0 \Rightarrow x \geq 3 \\ -x+3, x-3 < 0 \Rightarrow x < 3 \end{cases}$

$\Rightarrow |x-4| \begin{cases} x-4, x-4 \geq 0 \Rightarrow x \geq 4 \\ -x+4, x-4 < 0 \Rightarrow x < 4 \end{cases}$

$\Rightarrow |x-4| \begin{cases} x-4, x-4 \geq 0 \Rightarrow x \geq 4 \\ -x+4, x-4 < 0 \Rightarrow x < 4 \end{cases}$

$B = \{(x, y), (x, y) \in A^2\}$

A is bounded Set

1) Prove that B is bounded above

bounded above $|x - y| \leq M$

we have that A is bounded

$\inf(A) \leq x \leq \sup(A) - \epsilon$

$\inf(A) \leq y \leq \sup(A) - \epsilon$

$-\sup(A) \leq -y \leq -\inf(A) - (2)$

(1) - (2)

$\inf - \sup \leq x - y \leq \sup(A) - \inf(A)$

$-(\sup(A) - \inf(A)) \leq x - y \leq \sup(A) - \inf(A)$

$-M \leq x - y \leq M$

So $x - y \leq \sup(A) - \inf(A)$

So (B) is Bounded above

2) show that $\sup(B) = \sup(A) - \inf(A)$

characterization of $\sup(A)$

$\forall \epsilon > 0, \exists x_0 \in A, \sup(A) - \epsilon < x_0 < \sup(A)$

" of $\inf(A)$

$\forall \epsilon > 0, \exists x_0 \in A, \inf(A) < x_0 < \inf(A) + \epsilon$

$\sup(A) - \inf(A) < x_0 - y_0 < \sup(A) - \inf(A)$

we have $\sup(B) = \sup(A) - \inf(A)$

⁺ Définition de sup et inf d'un ensemble

borné supérieur : $x \leq M$
 borné inférieur : $x \geq m$
 borné : $m \leq x \leq M$

Caractérisation de $\sup A = M$

$$\begin{cases} \forall x \in A; x \leq M \\ \forall \varepsilon > 0, \exists x \in A, x > M - \varepsilon \end{cases}$$

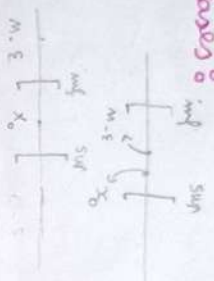
we have two cases :

case 01 :

$$M - \varepsilon > \inf A$$

case 02 :

$$M - \varepsilon < \inf A$$



Caractérisation de $\inf A = m$

$$\begin{cases} \forall x \in A; x \geq m \\ \forall \varepsilon > 0, \exists x \in A, x < m + \varepsilon \end{cases}$$

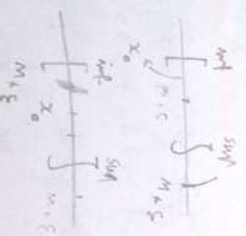
we have two cases :

case 01 :

$$m + \varepsilon > \sup A$$

case 02 :

$$m + \varepsilon < \sup A$$



Densité in $\mathbb{R}$: $\bar{Q} = \mathbb{R}$

$$\forall x \in \mathbb{R}, y \in \mathbb{R} (x < y) \Rightarrow \exists q \in \mathbb{Q},$$

$$x < q < y$$

Therefore : $\bar{Q} = \mathbb{R}$

Interval :

$$[a, b]; \forall a, b \in \mathbb{R}; a < b$$

$\Delta \mathbb{Q}, \mathbb{Z}, \mathbb{N} \Rightarrow$ aren't an interval

Law 5 :

$$[x] = \lfloor x \rfloor$$

$$[x + m] = [x] + m$$

$$[x] \leq x \leq [x] + 1$$

$$x^{-1} \leq [x] \leq x$$



$$\sup(A \cup B) = \max \{ \sup A, \sup B \}$$

$$\inf(A \cup B) = \min \{ \inf A, \inf B \}$$

$$\sup(A + B) = \sup A + \sup B$$

$$\inf(A + B) = \inf A + \inf B$$



$$(a + b)^m = \sum_{k=0}^m C_m^k \cdot a^k \cdot b^{m-k}$$

$$C_m^k = \frac{m!}{k!(m-k)!}$$

$$(x^n + y^n) = (x - y)(x^{n-1} + y^{n-1} + x^{n-2}y + \dots + y^{n-1})$$



$$\sum_{n=0}^{\infty} (ax)^n \Rightarrow T = \frac{2\pi}{\alpha}$$

Révision finale de chapitre 03

1/ Bounded sequences:

- bounded above $u_n \leq M$
- bounded below $u_n \geq m$
- bounded $m \leq u_n \leq M$ / $|u_n| \leq M$
- Unbounded $|u_n| > M$

2/ increasing / decreasing:

$$\left. \begin{array}{l} u_{n+1} - u_n \leq 0 \\ \frac{u_{n+1}}{u_n} \leq 1 \end{array} \right\} \text{decreasing}$$

↓
or monotone

$$\left. \begin{array}{l} u_{n+1} - u_n \geq 0 \\ \frac{u_{n+1}}{u_n} \geq 1 \end{array} \right\} \text{increasing}$$

↑

3/ Subsequences:

$$n \in \mathbb{N}^+ \nearrow$$

4/ convergent sequences:

u_n is C.V. for l :

$$\left\{ \begin{array}{l} \forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall m \in \mathbb{N}, m \geq n_0 \\ |u_m - l| < \varepsilon \end{array} \right.$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall m \in \mathbb{N}, m \geq n_0$$

$$-\varepsilon < u_m - l < \varepsilon \Rightarrow l - \varepsilon < u_m < l + \varepsilon$$

$$\Leftrightarrow \lim u_n = l \text{ (existence finite).}$$

5/ Convergent theorems:

Theorem 01:

- every convergent sequence is bounded

Theorem 02:

$$\lim u_n = 0 \Leftrightarrow \lim |u_n| = 0$$

$$\lim u_n = l \Rightarrow \lim |u_n - l| = |l - l| = 0$$

if u_n is bounded and v_n is C.V. so

$$\lim u_n \cdot v_n = 0$$

Theorem 03: Squeeze theorem

$$1) \exists m \in \mathbb{N}, \forall n \in \mathbb{N}, m \geq n_0 \Rightarrow v_n \leq u_n \leq w_n$$

$$2) \lim v_n = \lim w_n = l \text{ (finite)}$$

$$\text{then } \lim u_n = l$$

Theorem 04:

u_n bounded above + increasing = C.V

$$\text{and } \lim u_n = \sup \{u_n, n \in \mathbb{N}\}$$

u_n bounded below + decreasing = C.V

$$\text{and } \lim u_n = \inf \{u_n, n \in \mathbb{N}\}$$

6/ convergent with subsequence

Theorem:

if u_n is a C.V. sequence for l then

any of its subsequences also C.V.

properties:

$$\begin{array}{c} u_m \\ \swarrow \searrow \\ u_{2m} \quad u_{3m} \end{array}$$

$$\lim u_{2m} = \lim u_{3m} = l$$

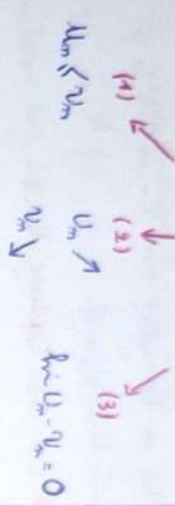
$$\Rightarrow u_n \text{ is C.V. and } \lim u_n = l$$

Theorem: Bolzano-Weierstrass

every bounded sequence has a C.V. sequence.

7/ adjacent sequence

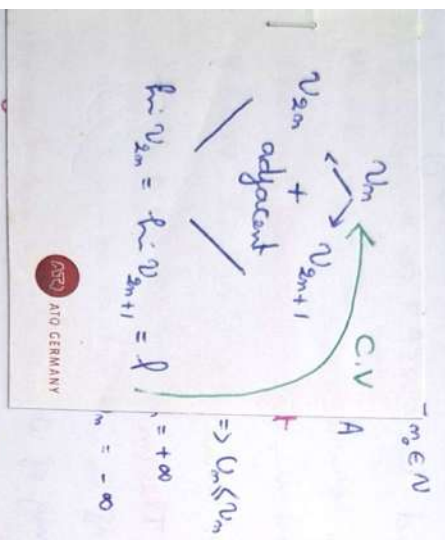
u_n and v_n are adjacent



8/ divergent sequence

We say that u_n is d.v.:

- 1) $\lim_{n \rightarrow \infty} u_n = +\infty \Leftrightarrow \forall A > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq n_0 \Rightarrow u_n > A$



- u_n is $\nearrow$ and x bounded above
- So $\lim u_n = +\infty$
- u_n is $\searrow$ and x bounded below
- So $\lim u_n = -\infty$

- $\lim u_n = +\infty / -\infty$ all subsequences of u_n are d.v.
- if two subsequences from u_n are C.V. to two different limits then u_n is d.v.

9/ Cauchy sequences

We say that u_n is the Cauchy Seq:

$$\forall \epsilon > 0, \exists n_0 \in \mathbb{N}, \forall p, q \in \mathbb{N}, p > q > n_0 \Rightarrow |u_p - u_q| < \epsilon$$

We say that u_n is not the Cauchy

$$\exists \epsilon > 0, \forall n_0 \in \mathbb{N}, \forall p, q \in \mathbb{N}, p > q > n_0 \Rightarrow |u_p - u_q| > \epsilon$$

Theorem:

u_n is the Cauchy $\rightarrow$ bounded
 u_n is C.V. $\Leftrightarrow u_n$ is the Cauchy

10/ Recursive Sequences

$$u_{n+1} = f(u_n)$$

subsequences

u_{2n} and u_{2n+1}

monotone $\neq$ direction

if u_{2n} and u_{2n+1} are

bounded then 2 limits

are C.V. for f so $f(l) = l$

So $\lim u_{2n} = \lim u_{2n+1} = l$

$\lim u_n = l \Rightarrow u_n$ is C.V.

u_n is monotone

$$u_1 - u_2 > 0 \nearrow$$

C.V. + bounded

$$u_n \leq M$$

$$u_1 - u_2 < 0 \searrow$$

C.V. + b. below

$$u_n \geq m$$

Theorem: Bolzano-Weierstrass

every bounded sequence has a C.V. subsequence.

ϵ / adjacent sequence

u_n and v_n are adjacent

$$\begin{array}{ccc} (1) \swarrow & \downarrow & \searrow (3) \\ u_n & & v_n \\ (2) \nearrow & & \nwarrow \\ u_n & & v_n \end{array} \quad \lim u_n - v_n = 0$$

8 / divergent sequence

we say that u_n is d.v.

$$1) \lim_{n \rightarrow \infty} u_n = +\infty \Leftrightarrow \forall A > 0, \exists m_0 \in \mathbb{N} \forall n \in \mathbb{N}, n \geq m_0 \Rightarrow u_n > A$$

$$\lim_{n \rightarrow \infty} u_n = -\infty \Leftrightarrow \forall A > 0, \exists m_0 \in \mathbb{N} \forall n \in \mathbb{N}, n \geq m_0 \Rightarrow u_n < -A$$

Theorem: comparison test

$\exists k \in \mathbb{N}, \forall n \in \mathbb{N} (n \geq k) \Rightarrow u_n \leq v_n$

$$\lim u_n = +\infty \Rightarrow \lim v_n = +\infty$$

$$\lim u_n = -\infty \Rightarrow \lim v_n = -\infty$$

divergent result

u_n is $\nearrow$ and x bounded above

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = +\infty$$

u_n is $\searrow$ and x bounded below

$$\Rightarrow \lim_{n \rightarrow \infty} u_n = -\infty$$

$\lim u_n = +\infty / -\infty$ all subsequences of u_n are d.v.

if two subsequences from u_n are C.V. to two different limits then u_n is d.v.

9 / Cauchy Sequences

we say that u_n is the Cauchy Seq:

$$\forall \epsilon > 0, \exists m_0 \in \mathbb{N}, \forall p, q \in \mathbb{N}, p, q > m_0 \Rightarrow |u_p - u_q| < \epsilon$$

$$\Rightarrow |u_p - u_q| < \epsilon$$

we say that u_n is not the Cauchy

$$\exists \epsilon > 0, \forall m_0 \in \mathbb{N}, \forall p, q \in \mathbb{N}, p, q > m_0$$

$$\Delta |u_p - u_q| > \epsilon$$

Theorem:

u_n is the Cauchy $\rightarrow$ bounded

u_n is C.V. $\Leftrightarrow u_n$ is the Cauchy

10 / Recursive Sequences

$$u_{n+1} = f(u_n)$$

subsequences

$$u_{2m} \text{ and } u_{2m+1}$$

monotone $\neq$ direction

$$u_{2m} \text{ and } u_{2m+1} \text{ or } u_{2m+1}$$

bounded then 2 limits

$$\text{are C.V. to } l \text{ so } f(l) = l$$

$$\lim u_{2m} = \lim u_{2m+1} = l$$

$$\lim u_n = l \Rightarrow u_n \text{ is C.V.}$$

u_n is monotone

$$u_1 - u_2 > 0 \nearrow$$

C.V. + bounded

$$u_n \leq M$$

$$u_1 - u_2 < 0 \searrow$$

C.V. + bounded

$$u_n \geq m$$

Solution Series + exercises

1/ Solution ex of cours

$$1) U_n = \left| \frac{(-1)^n n + 26}{\sqrt{n^2 + 2}} \right|$$

$$U_n = \frac{|(-1)^n n + 26|}{\sqrt{n^2 + 2}}$$

$$|a+b| \leq |a| + |b|$$

$$|(-1)^n n + 26| \leq |(-1)^n n| + |26|$$

$$\frac{|(-1)^n n + 26|}{\sqrt{n^2 + 2}} \leq \frac{|(-1)^n n| + 26}{\sqrt{n^2 + 2}}$$

$$\frac{|(-1)^n n + 26|}{\sqrt{n^2 + 2}} \leq \frac{n + 26}{\sqrt{n^2 + 2}} = \frac{n}{\sqrt{n^2 + 2}} + \frac{26}{\sqrt{n^2 + 2}}$$

$$\bullet n^2 + 2 \geq n^2 \Rightarrow \sqrt{n^2 + 2} \geq \sqrt{n^2} = n$$

$$\Rightarrow \frac{n}{\sqrt{n^2 + 2}} \leq \frac{n}{n} = 1$$

$$\bullet n^2 + 2 \geq 2 \Rightarrow \sqrt{n^2 + 2} \geq \sqrt{2}$$

$$\frac{1}{\sqrt{n^2 + 2}} \leq \frac{1}{\sqrt{2}}$$

$$\frac{26}{\sqrt{n^2 + 2}} \leq \frac{26}{\sqrt{2}} \quad (3)$$

$$1+2 \Rightarrow \frac{n}{\sqrt{n^2 + 2}} \leq 1 + \frac{26}{\sqrt{2}}$$

$$|U_n| \leq M$$

$$\exists M = 1 + \frac{26}{\sqrt{2}} \in \mathbb{R}, \forall n \in \mathbb{N}, |U_n| \leq M$$

So U_n is bounded.

$$2) U_n = \sum_{k=1}^n \frac{1}{k^2}, \forall n \in \mathbb{N}$$

$$U_{n+1} - U_n = \sum_{k=1}^{n+1} \frac{1}{k^2} - \sum_{k=1}^n \frac{1}{k^2}$$

$$= \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} + \frac{1}{(n+1)^2} \right] -$$

$$\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} \right] = \frac{1}{(n+1)^2} > 0$$

$U_{n+1} - U_n > 0$ So U_n is increasing

$$3) U_n = \frac{2^n}{n!}$$

$$\frac{U_{n+1}}{U_n} = \frac{2^{(n+1)}}{(n+1)!} = \frac{2^{n+2}}{(n+1)!} \cdot \frac{n!}{2^n}$$

$$= \frac{2^{n+2}}{(n+1)n!} \cdot \frac{n!}{2^n} = \frac{2^{n+2}}{(n+1)n!} \cdot \frac{1}{2^n}$$

$$\text{So: } U_{n+1} < U_n$$

$$U_n \text{ is decreasing}$$

$$= 2 \cdot \frac{1}{2^n}$$

$$\Rightarrow \frac{1}{n} \Rightarrow 0 < \frac{1}{n} < 1$$

$$4) U_n = e^{2n + \frac{1}{n}}$$

$$\frac{U_{n+1}}{U_n} = \frac{e^{2(n+1) + \frac{1}{n+1}}}{e^{2n + \frac{1}{n}}}$$

$$= e^{2n + 2 + \frac{1}{n+1} - 2n - \frac{1}{n}}$$

$$= e^{2 + \frac{1}{n(n+1)}}$$

$$= e^{2 + \frac{1}{n(n+1)}}$$

5) Prove that $\lim_{n \rightarrow \infty} U_n = \frac{1}{3}$ (obvious)

7 $U_n = \frac{m+1}{3m+2}$

by definition

a $\forall \epsilon > 0, \exists n_0 \in \mathbb{N}, \forall m \in \mathbb{N}, m \geq n_0$

$|U_n - \frac{1}{3}| < \epsilon$

$$\Rightarrow \left| \frac{m+1}{3m+2} - \frac{1}{3} \right| = \left| \frac{3m+3-3m-2}{9m+6} \right|$$

$$= \frac{1}{9m+6} < \epsilon$$

$$= 3m+6 > \frac{1}{\epsilon}$$

$$= m > \frac{1}{9\epsilon} - 6$$

we take $n_0 = \left\lceil \frac{1}{9\epsilon} - 6 \right\rceil + 1$ so

$$\lim_{n \rightarrow \infty} U_n = \frac{1}{3}$$

6/ $U_n = 0 \rightarrow U_n = \frac{1}{n^2}$

by definition

$\forall \epsilon > 0, \exists n_0 \in \mathbb{N}, \forall m \in \mathbb{N}, m \geq n_0$

$|U_n - 0| < \epsilon$

$$\left| \frac{1}{m^2} \right| < \epsilon$$

$$m^2 > \frac{1}{\epsilon}$$

$$m > \sqrt{\frac{1}{\epsilon}} \Rightarrow \text{we take } n_0 = \left\lceil \sqrt{\frac{1}{\epsilon}} \right\rceil + 1$$

then $\lim_{n \rightarrow \infty} U_n = 0$

6/ $U_n = \frac{\sin m}{m}$ let $\begin{cases} U_n = \sin m \\ v_m = \frac{1}{m} \end{cases}$

$|U_n| = |\sin m| \leq 1$: bounded

and $\lim_{n \rightarrow \infty} v_m = 0$ so

$$\lim_{n \rightarrow \infty} U_n \cdot v_m = \lim_{n \rightarrow \infty} U_n = 0$$

(2)

8/ $U_n = \sum_{k=0}^n \frac{1}{m^2+k}$

we have : $0 \leq k \leq 1$

Cauchy's $m^2 \leq m^2+k \leq 1+m^2$

by $\frac{1}{1+m^2} \leq \frac{1}{m^2+k} \leq \frac{1}{m^2}$

$$\sum_{k=0}^n \frac{1}{1+m^2} \leq \sum_{k=0}^n \frac{1}{m^2+k} \leq \sum_{k=0}^n \frac{1}{m^2}$$

$$\frac{n+1}{1+m^2} \leq U_n \leq \frac{n+1}{m^2}$$

$$U_n \leq U_m \leq U_m$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} U_n = 0$$

So by the squeeze theorem we get

$$\lim_{n \rightarrow \infty} U_n = 0 \Rightarrow U_n \text{ is } \mathcal{O}(1/n)$$

9/ $U_n = \sum_{k=0}^n \frac{1}{m+k}$

$$U_{n+1} - U_n = \sum_{k=0}^{n+1} \frac{1}{m+k} - \sum_{k=0}^n \frac{1}{m+k}$$

$$= \left[\frac{1}{m+2} + \frac{1}{m+3} + \frac{1}{2m} + \frac{1}{2m+1} + \frac{1}{2m+3} \right]$$

$$- \left[\frac{1}{m+1} + \frac{1}{m+2} + \frac{1}{2m} \right]$$

$$= \left[\frac{1}{2m+1} + \frac{1}{2m+3} - \frac{1}{m+1} \right] > 0$$

then U_n is increasing

$$m+1 \geq m+1 \rightarrow \frac{1}{m+1} \leq \frac{1}{m+1}$$

$$m+2 \geq m+2 \rightarrow \frac{1}{m+2} \leq \frac{1}{m+2}$$

$$m+3 \geq m+3 \rightarrow \frac{1}{m+3} \leq \frac{1}{m+3}$$

!

$$m+n \geq m+1 \rightarrow \frac{1}{m+n} \leq \frac{1}{m+1}$$

$$\frac{1}{m+1} + \frac{1}{m+2} + \frac{1}{m+3} \leq \frac{1}{m+1} + \frac{1}{m+1} + \frac{1}{m+1}$$

$$\sum_{k=1}^n \frac{1}{m+k} < m \left(\frac{1}{m+1} \right)$$

$\frac{m}{m+1} < 1$ So U_n is bounded above by 1 then U_n is CV

10/ $U_n = \frac{1}{2^n}$

Let $U_n = \frac{1}{n} \rightarrow U_n = U_{2^n} / 2^n \nearrow$

So U_n is a subsequence from U_n then $\lim U_n = 0 \rightarrow U_n$ is C.V

11/ adjacent seq :

$$U_n = \sum_{k=1}^n \frac{1}{k^2} \text{ and } U_n = U_{n-1} + \frac{1}{n^2}$$

1) $U_n < U_m$

2) $U_{m+1} - U_m = \sum_{k=1}^{m+1} \frac{1}{k^2} - \sum_{k=1}^m \frac{1}{k^2}$

$$= \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{m^2} + \frac{1}{(m+1)^2} -$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{m^2}$$

$$= \frac{1}{(m+1)^2} > 0 \Rightarrow U_n \text{ is } \nearrow$$

3) $U_{m+1} - U_m = \left[\sum_{k=0}^{m+1} \frac{1}{k^2} + \frac{1}{m+1} \right] -$

$$\left[\sum_{k=0}^m \frac{1}{k^2} + \frac{1}{m} \right]$$

$$= \frac{1}{(m+1)^2} + \frac{1}{m+1} - \frac{1}{m}$$

$$= - \frac{m+1}{(m+1)^2(m^2+m)} < 0 \Rightarrow U_n \text{ is } \searrow$$

• $\lim U_n - U_m = 0$

So U_n and U_m are adjacent

12/ prove that $\lim \sqrt{m} + 1 = +\infty$

by definition :

$$\forall A > 0, \exists m_0 \in \mathbb{N}, \forall m \in \mathbb{N}, m > m_0$$

$$\Rightarrow U_n > A$$

$$\sqrt{m} + 1 > A$$

$$\sqrt{m} > A - 1$$

$$m > (A-1)^2 \rightarrow m_0 = [(A-1)^2] + 1$$

then $\lim U_n = +\infty$

13/ prove that $\lim -m^3 + 2 = -\infty$

by definition :

$$\forall A > 0, \exists m_0 \in \mathbb{N}, \forall m \in \mathbb{N}, m > m_0$$

$$\Rightarrow U_m < -A$$

$$-m^3 + 2 < -A$$

$$m^3 > A + 2 \rightarrow m > (A+2)^{1/3}$$

$$m_0 = [(A+2)^{1/3}] + 1 \text{ then } \lim U_m = -$$

14/ $U_m = \frac{m^2}{2} + \sin m$

$$-1 < \sin m < 1$$

$$\frac{m^2}{2} - 1 < \frac{m^2}{2} + \sin m < \frac{m^2}{2} + 1$$

Let $U_m = \frac{m^2}{2} - 1 \Rightarrow U_m < U_m$

We have $\lim U_m = \lim V_m = +\infty$

by comparison test $\lim U_m = +\infty$

So U_m is d.v

(3)

15/ $U_n = \frac{n+1}{n}$

Show that U_n is the Cauchy

by definition:

$\forall \epsilon > 0, \exists m_0 \in \mathbb{N}, \forall p, q \in \mathbb{N}$

$p > q > m_0 \Rightarrow |U_p - U_q| < \epsilon$

$|U_p - U_q| = \left| \frac{p+1}{p} - \frac{q+1}{q} \right|$

$= \left| 1 + \frac{1}{p} - 1 - \frac{1}{q} \right|$

$= \left| \frac{1}{q} - \frac{1}{p} \right|$

$\left| \frac{1}{q} - \frac{1}{p} \right| \leq \left| \frac{1}{p} \right| + \left| \frac{1}{q} \right| = \frac{1}{p} + \frac{1}{q}$

we have: $p > q$

$\Rightarrow \frac{1}{p} < \frac{1}{q} \Rightarrow \frac{1}{p} + \frac{1}{q} < \frac{2}{q}$

$\frac{2}{q} < \epsilon \rightarrow q > \frac{2}{\epsilon} \rightarrow m_0 = \left\lceil \frac{2}{\epsilon} \right\rceil + 1$

then U_n is the Cauchy.

16/ EXO d'examen 2023

we have: $U_0 = 2$

$U_{n+1} = \frac{2}{1+U_n}$

1) by induction prove that:

$\frac{1}{2} \leq U_n \leq 2$

for $P(m) \Rightarrow \frac{1}{2} \leq 2 \leq 2$ ✓

we assume that $P(m)$ is true and

we prove that $P(m+1)$ is also true

we have:

$\frac{1}{2} \leq U_m \leq 2$

(4) $\frac{2}{2} \leq U_{m+1} \leq 3$

$\frac{2}{3} \leq \frac{2}{U_{n+1}} \leq \frac{4}{3}$ ✓

$\frac{1}{2} \leq \frac{2}{3} \leq U_n \leq \frac{4}{3} \leq 2$

by induction $P(m)$ is true.

2) is the U_n monotone?

$f(x) = \frac{2}{x+1} \rightarrow f'(x) = -\frac{2}{(1+x)^2} < 0$

Since U_n is decreasing then U_n is not monotone.

3) the nature of subsequences:

$\{U_{2n}\}_n \Rightarrow$ we have $f(x)$ is $\searrow$ so the

$\{U_{2n+1}\}_n$ Subsequences are monotone

with $\neq$ direction and are

have $\frac{1}{2} \leq U_n \leq 2$ as U_n is bounded

then all of subsequences are bounded

So: U_{2n} and U_{2n+1} are monotone

and bounded $\Rightarrow$ they are C.V. to f

4) is (U_n) C.V.?

we calculate f :

$f \circ f(f) = f \Rightarrow f(f(f)) = f$

So U_n is $\frac{2}{1+f(f)} = f$

C.V. and

$\lim U_n = f$

$\frac{2}{1+\frac{2}{1+f}} = f$

$f^2 + 2 - 2 = 0$

$\lim U_{2n} = \lim U_{2n+1} = f$

1.60
-2.80

$$\text{Let } E = \{U_m : m \in \mathbb{N}\}$$

determine $\sup, \inf, \min, \max$

we have U_{2n} and U_{2n+1} are monotone:

$$U_2 - U_0 = -\frac{4}{5} < 0$$

So U_{2n} is $\searrow$ and U_{2n+1} is $\nearrow$

$$U_1 < U_3 < U_5 < U_7 \dots < U_{2n+1} <$$

$$U_{2n} < U_{2n-2} < U_{2n-4} < U_{2n-6} < U_{2n-8} < U_{2n-10} < \dots < U_2 < U_0$$

$$\text{So: } \sup E = \max E = U_0 = 2$$

$$\inf E = \min E = U_1 = \frac{2}{3}$$

Exo 2 - examen 2023

we have: $\int U_0 = 1$

$$\int U_{n+1} = \frac{U_n + 1}{U_n + 3}$$

1/ by induction $0 \leq U_n \leq 1$

2/ U_n is monotone?

$$f(x) = \frac{x+1}{x+3}$$

$$f'(x) = \frac{2}{(x+3)^2} > 0 \text{ since } U_n$$

is $\nearrow$ then is monotone

$$U_1 - U_0 = -\frac{1}{2} < 0 \text{ so it's } \searrow$$

3/ deduce that U_n is C.V

we have by qst 1 U_n is bounded below by 0 and it's

So it's C.V and the li-

$$f = f(1) \rightarrow \frac{f+1}{f+3} = f$$

$$\Rightarrow f^2 + 2f - 1 = 0$$

$$\begin{cases} f_1 = -1 + \sqrt{2} \\ f_2 = -1 - \sqrt{2} \end{cases}$$

$$\text{So } \lim U_n = -1 + \sqrt{2}$$

$$\text{Let } E = \{U_n : n \in \mathbb{N}\}$$

U_n is $\searrow$ so:

$$U_0 > U_1 > U_2 > U_3 > U_4 > \dots > U_n$$

$$\sup E = \max E = U_0 = 1$$

and we have U_n is $\searrow$ and bounded

below by 0 $\Rightarrow \inf E = \lim U_n = -1 + \sqrt{2}$

$$-1 + \sqrt{2} \in E \text{ so } \min E$$

Exo 3 - examen 2023

we have: $\int U_0 = 2$

$$\int U_{n+1} = 5 - \frac{4}{U_n}$$

1/ by induction: $1 < U_n < 4$

2/ prove that U_n is C.V

$$f(x) = 5 - \frac{4}{x}$$

$$f'(x) = \frac{4}{x^2} > 0 \text{ so it's monotone}$$

$$U_1 - U_0 = 5 - \frac{4}{2} - 2 = 1 > 0 \text{ so it's } \nearrow$$

since U_n bounded above by 4 so it's

C.V and $\lim U_n = 4$

$$f = f(4) \rightarrow f^2 + 5f - 4 = 0$$

$$\text{So } \lim U_n = 4$$

Let $E = \{U_n, n \in \mathbb{N}\}$

determine $\sup, \inf, \max, \min$

• U_n is $\uparrow$ so:

$$U_0 \leq U_1 \leq U_2 \leq \dots \leq U_n$$

So: $\sup E = \max E = U_0 = 2$

• U_n is $\uparrow$ so: and bounded

above by 4 so: $\inf E = \min U_n = 1$
 $\min E = 1,$

$$\lim_{n \rightarrow \infty} (f(n))^{g(n)} = 1^4$$

$$= \lim_{n \rightarrow \infty} [f(n)^{-1}]^{g(n)}$$

Revision guide chapter 4

1) differentiability at a point:

We say that f is differentiable at x_0 if:

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \in \mathbb{R}$$

Then the limit is called the derivative of f at x_0 and

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

2) Diff on the right and left:

$$\text{Right: } \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'_+(x_0)$$

$$\text{Left: } \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'_-(x_0)$$

$$\text{diff on } x_0 \Rightarrow f'_-(x_0) = f'_+(x_0) = f'(x_0)$$

Δ mult diff if:

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \infty \text{ or } \infty$$

3) Theorem: (chain rule)

$$h(x) = f[g(x)]$$

$$h'(x) = [f(g(x))]'$$

$$h'(x) = g'(x) \cdot f'(g(x)).$$

For ex we have:

$$h(x) = \cos(x^2)$$

$$h'(x) = (x^2)' \cdot \cos'(x^2)$$

$$h'(x) = 2x \cdot (-\sin(x^2))$$

$$h'(x) = -2x \sin(x^2).$$

4) derivative of an inverse function:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

5) Rolle's theorem:

Let $f: [a, b] \rightarrow \mathbb{R}$ be a function

if:

• f is defined and continuous on $[a, b]$

• f is differentiable on $]a, b[$

• $f(a) = f(b)$

then: $\exists c \in]a, b[$; $f'(c) = 0$

6) Mean value theorem:

Let $f: [a, b] \rightarrow \mathbb{R}$ be a function if:

• f is continuous on $[a, b]$

• f is diff on $]a, b[$

then $\exists c \in]a, b[$: $f'(c) = \frac{f(b) - f(a)}{b - a}$

$$f(b) - f(a) = f'(c)(b - a).$$

$$\begin{cases}
 f^{(0)} = f \\
 f^{(1)} = f' \\
 f^{(2)} = f'' \\
 \vdots \\
 f^{(n-1)} = f^{(n-1)} \\
 f^{(n)} = f^{(n)}
 \end{cases}$$

$C^n \rightarrow f^{(n)}$
 $\rightarrow f^{(n)}(x_0)$

Let $f(x) = \frac{1}{x+1}$

determine $f^{(n)}(x)$.

The function f is differentiable

on D_f :

$$f^{(0)} = f(x) = \frac{1}{x+1}$$

$$f^{(1)} = f'(x) = -\frac{1}{(x+1)^2}$$

$$f^{(2)} = f''(x) = \frac{2(x+1)}{(x+1)^4} = \frac{2}{(x+1)^3}$$

$$f^{(3)} = f'''(x) = \frac{-2 \cdot 3(x+1)^2}{(x+1)^6} = -\frac{6}{(x+1)^4}$$

$$\vdots$$

$$f^{(n)} = \frac{(-1)^n n!}{(x+1)^{n+1}} \quad \forall n \in \mathbb{N}$$

now by induction we will show $P(m)$:

Set 1: for $m=0$

$$f^{(0)}(x) = \frac{1}{x+1} \Rightarrow \frac{(-1)^0 (0)!}{(x+1)^{0+1}} = \frac{1}{x+1}$$

so $P(0)$ is true

Set 2: $P(m+1) \Rightarrow f^{(m+1)}(x) = \frac{(-1)^{m+1} (m+1)!}{(x+1)^{m+2}}$

$$\begin{aligned}
 f^{(m+1)}(x) &= \left(f^{(m)}(x) \right)' \\
 &= \left[\frac{(-1)^m m!}{(x+1)^{m+1}} \right]' = \frac{(-1)^m m! \cdot (m+1) \cdot \frac{1}{(x+1)^{m+2}}}{(x+1)^{2m+2}}
 \end{aligned}$$

$P(m+1)$ is true then $= \frac{(-1)^{m+1} \cdot (m+1)!}{(x+1)^{m+2}}$

$P(m)$ also true.

8) function of class C^n

class C^n if:
 f is n -times diff
 $f^{(n)}$ is continuous

9) Leibniz formula:

$$(f \cdot g)^{(n)}(x) = \sum_{k=0}^n C_n^k \cdot f^{(k)}(x) \cdot g^{(n-k)}(x)$$

where $C_n^k = \frac{n!}{(n-k)! \cdot k!}$

4) Successive derivatives:

$$\begin{cases} f^{(0)} = f \\ f^{(k+1)} = (f^{(k)})' \end{cases}$$

$$\lim_{x \rightarrow x_0} \frac{f'(x) - f'(x_0)}{x - x_0} = f''(x_0)$$

methode:

$$\text{dét } f(x) = \frac{1}{x+1}$$

determine $f^{(n)}(x)$.

The function f is differentiable

on D_f :

$$f^{(0)} = f(x) = \frac{1}{x+1}$$

$$f^{(1)} = f'(x) = -\frac{1}{(x+1)^2}$$

$$f^{(2)} = f''(x) = \frac{2(x+1)}{(x+1)^4} = \frac{2}{(x+1)^3}$$

$$f^{(3)} = f'''(x) = \frac{-2 \times 3(x+1)^2}{(x+1)^6} = -\frac{1 \times 2 \times 3}{(x+1)^4}$$

$$\vdots$$

$$f^{(n)} = \frac{(-1)^n n!}{(x+1)^{n+1}} \quad \forall n \in \mathbb{N}$$

now by induction we will show

$P(n)$:

St 1: for $n=0$

$$f^{(0)}(x) = \frac{1}{x+1} \Rightarrow \frac{(-1)^0 (0)!}{(x+1)^{0+1}} = \frac{1}{x+1}$$

So $P(0)$ is true

$$\text{St 2: } P(n+1) \Rightarrow f^{(n+1)}(x) = \frac{(-1)^{n+1} (n+1)!}{(x+1)^{n+2}}$$

$$f^{(n+1)}(x) = (f^{(n)}(x))'$$

$$= \left[\frac{(-1)^n n!}{(x+1)^{n+1}} \right]' = \frac{(-1)^n n!}{(x+1)^{2n+1}} \cdot \frac{(-n-1)}{(x+1)^2} = \frac{(-1)^{n+1} (n+1)!}{(x+1)^{n+2}}$$

$P(n+1)$ is true then $\frac{(-1)^{n+1} (n+1)!}{(x+1)^{n+2}}$

$P(n)$ also true, $\frac{(-1)^n n!}{(x+1)^{n+1}}$

2) function of class C^m

class C^m if:

f is m -times diff

$\circ f^{(m)}$ is continuous

3) Leibniz formula:

$$(f \cdot g)^{(m)}(x) = \sum_{k=0}^m C_m^k \cdot f^{(k)}(x) \cdot g^{(m-k)}(x)$$

$$\text{where } C_m^k = \frac{m!}{(m-k)! k!}$$